

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Traversal Challenge

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	Assessment:	Assessment Tool 2 – Traversal Challenge
Weightage:	30% of CIA	Total Marks:	100
CO3 Statement:	Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, B-Trees, Red-Black Trees, and Splay Trees.		
Student Name:	Murshidha M	Reg. No.:	192571007
Section / Batch:	A	Date:	03/08/2026

Code of Conduct

I Murshidha M (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Murshidha

Instructions

- This assessment tool contains 5 Traversal Challenge questions, Total: 100 Marks.
- Students can use any online/offline Traversal Challenge. Demonstrate tree.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

Section / Topic	Focus	Q Nos	Marks
Tree Traversals	Traversal types, In order, Post order and Pre order	5	100
GRAND TOTAL:			100 Marks

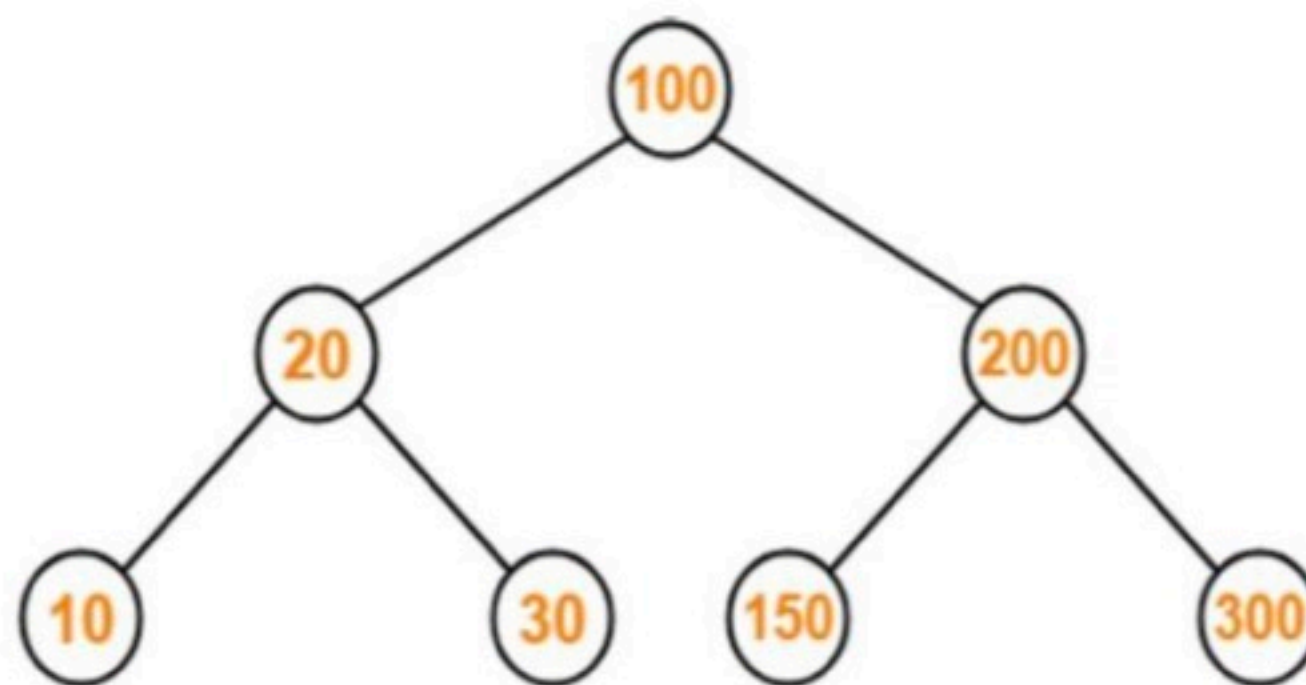
Traversal Challenge

Rubrics for Calculation

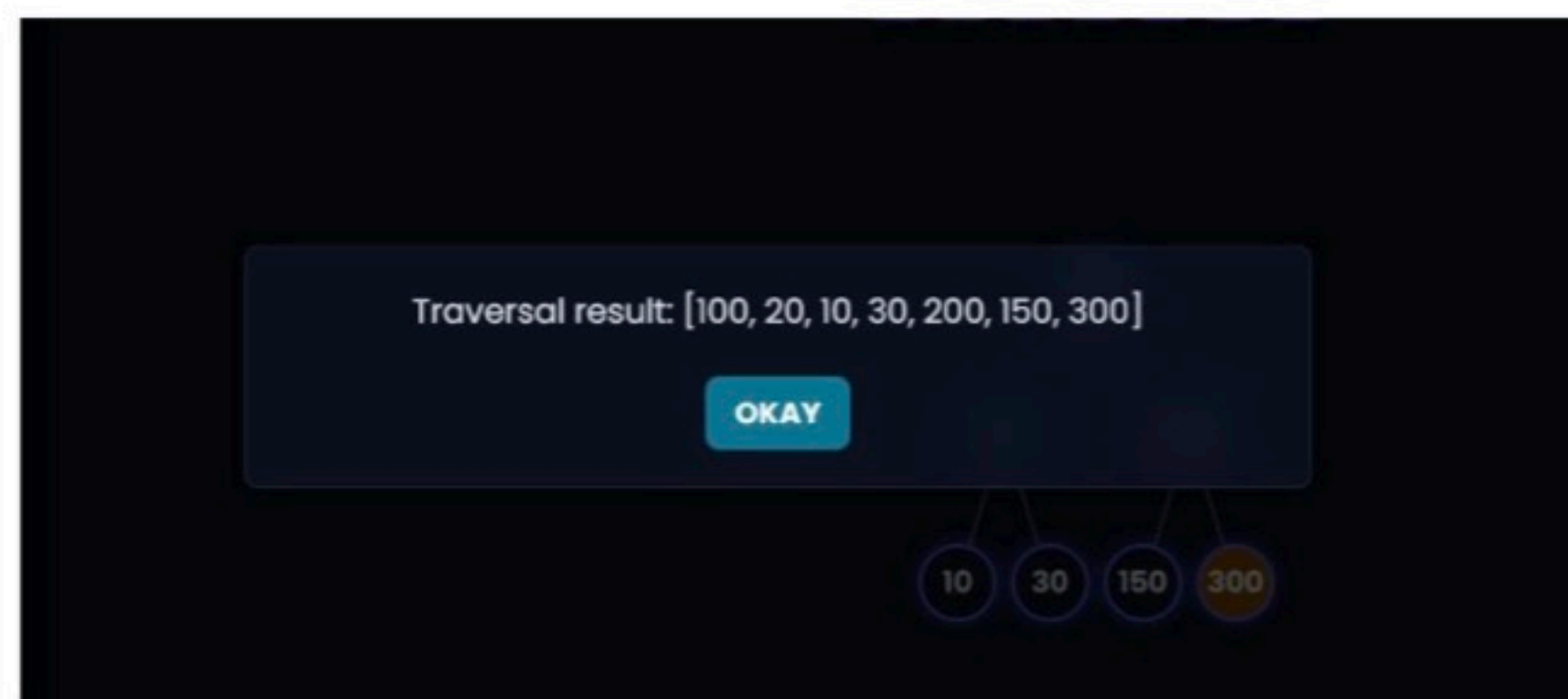
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	20	Demonstrates thorough understanding of all core Data Structures concepts; answers accurately reflect deep knowledge	Shows good understanding with minor conceptual gaps	Basic understanding; several misconceptions present	Limited understanding; major concepts misunderstood
Traversal Accuracy	20	Correctly completes all tree traversals (Preorder, Inorder, Postorder, Level Order) without errors.	Completes most traversals correctly with one minor error.	Correctly performs only some traversals with multiple errors.	Unable to perform most traversals correctly.
Selection of Appropriate Traversal	30	Correctly selects the appropriate traversal for every given problem scenario.	Selects the correct traversal for most scenarios.	Selects appropriate traversal for only a few scenarios.	Frequently selects incorrect traversal methods.
Completion & Time Efficiency	20	Completes all challenges within the allotted time while maintaining accuracy.	Completes most challenges within the time limit.	Completes only part of the challenge or exceeds the time limit slightly.	Unable to complete the challenge within the allotted time.
Evidence Submission	10	Uploads complete screenshots showing successful completion, score, and student details.	Uploads screenshots with minor missing information.	Uploads incomplete or unclear screenshots.	No valid evidence submitted.

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

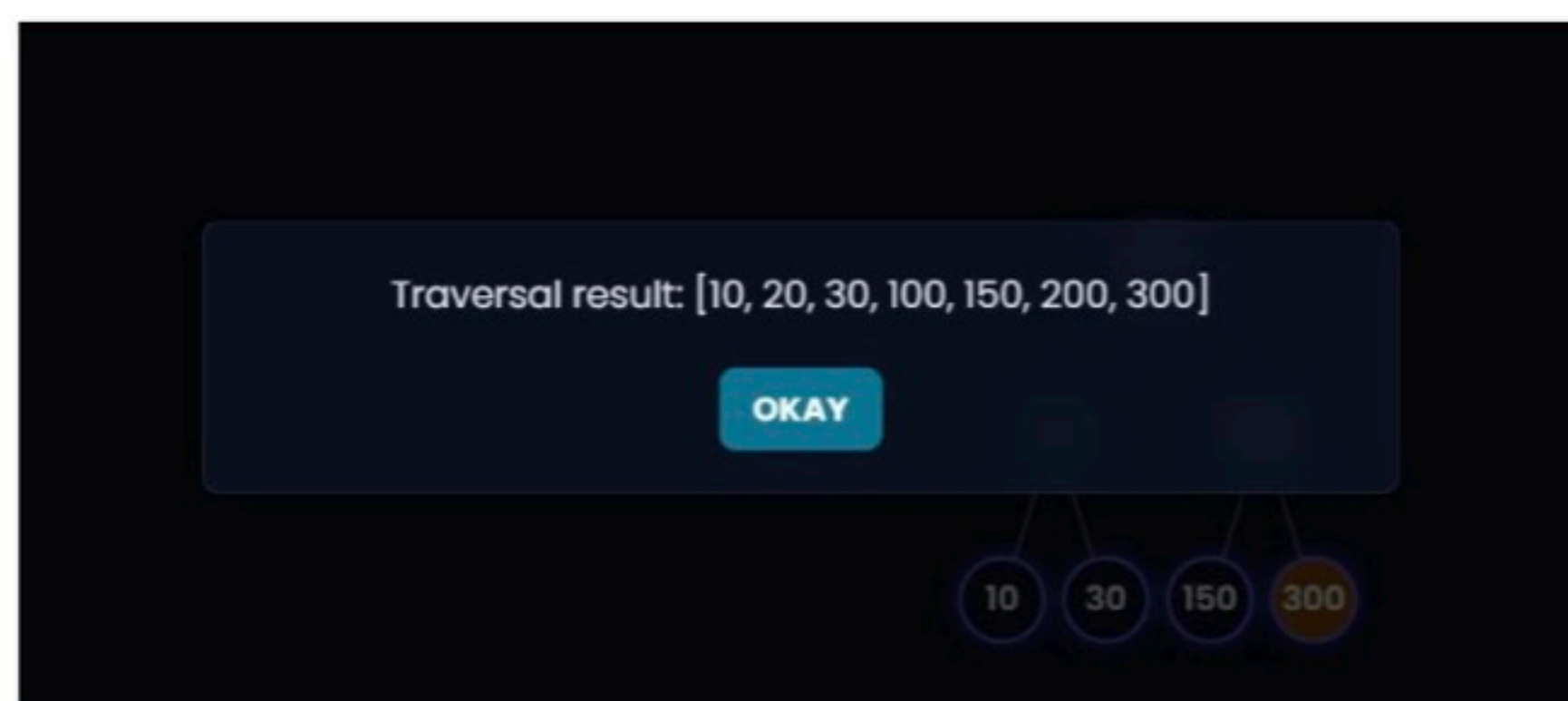
1. Write the traversal sequences for following trees



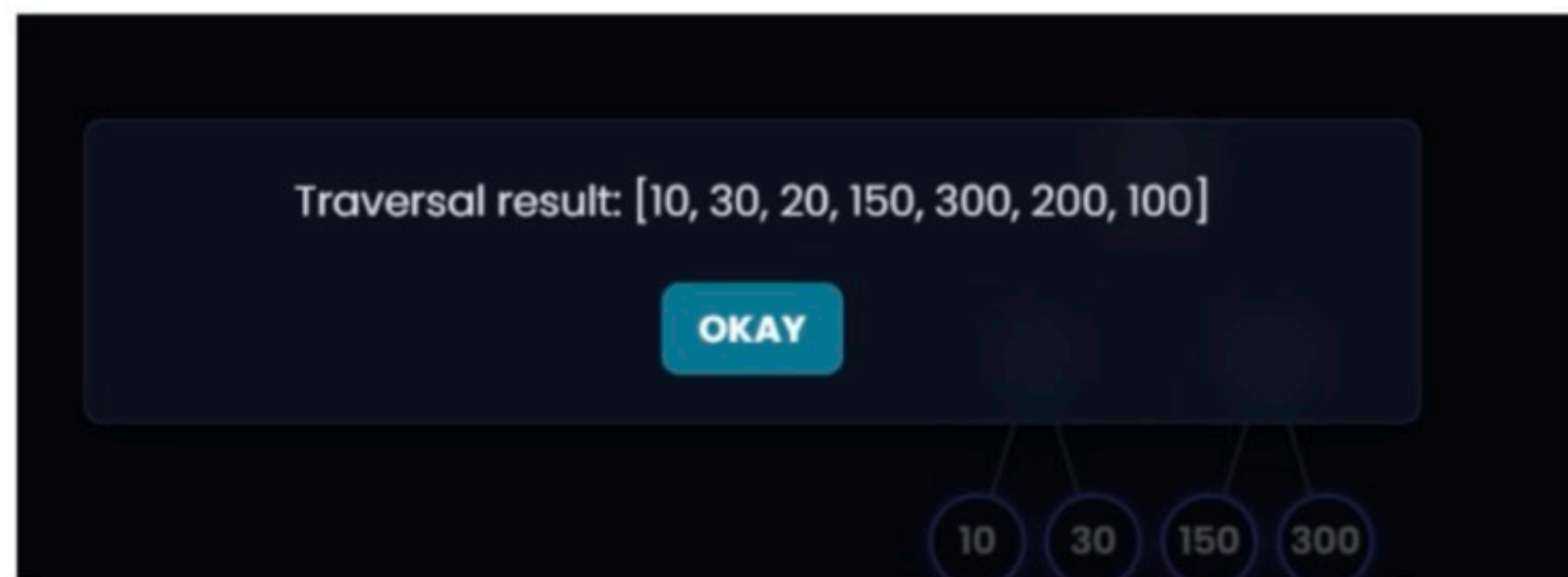
PREORDER



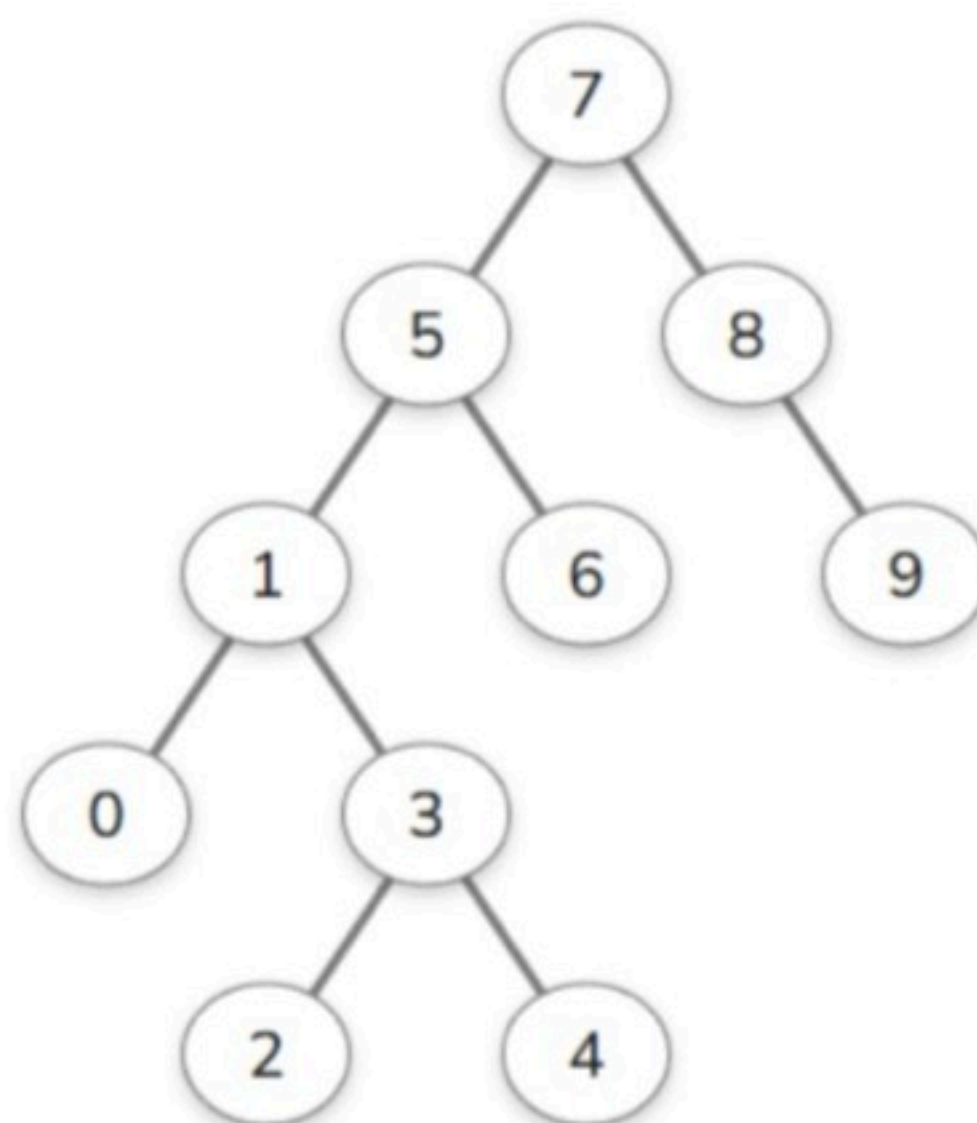
INORDER



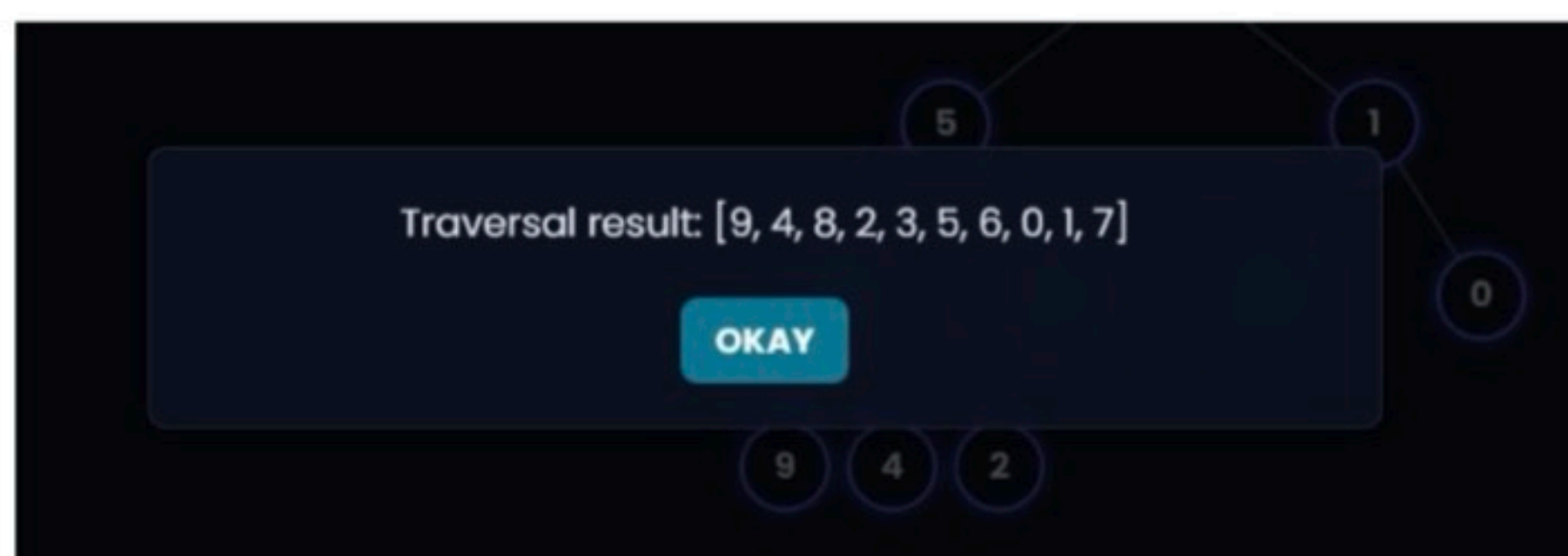
POSTORDER



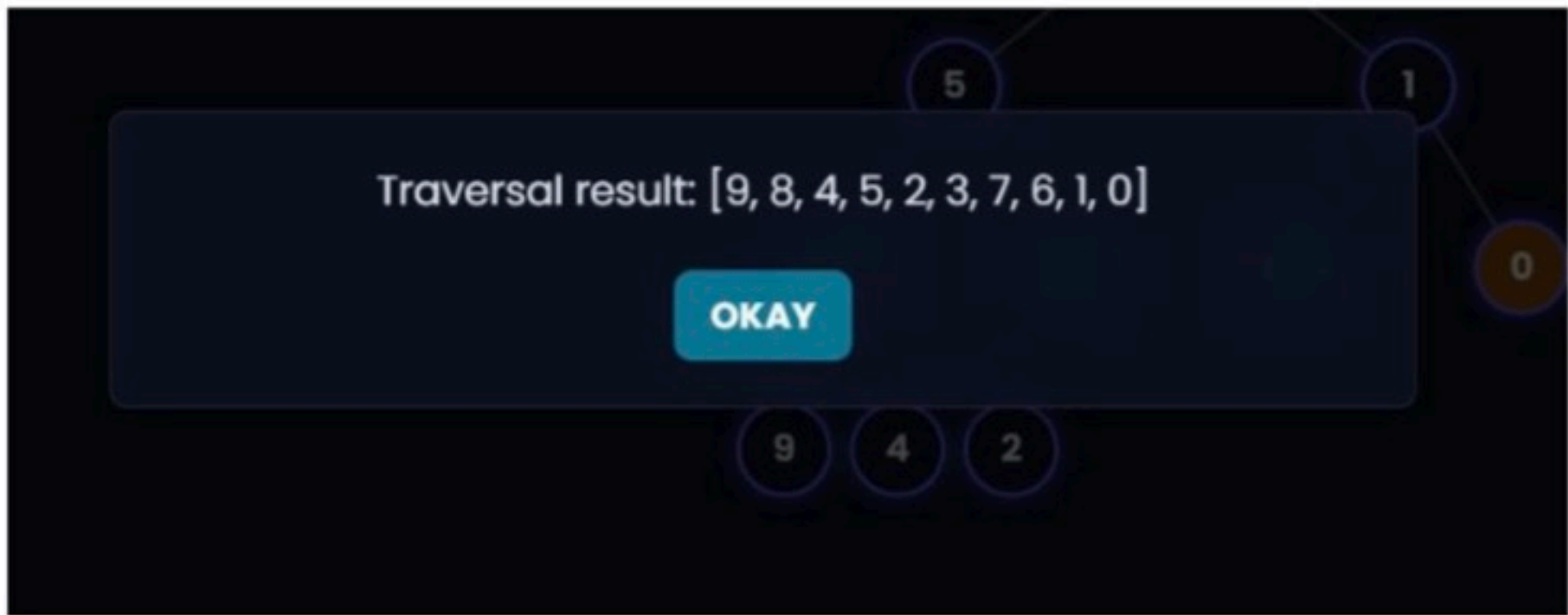
2. Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?



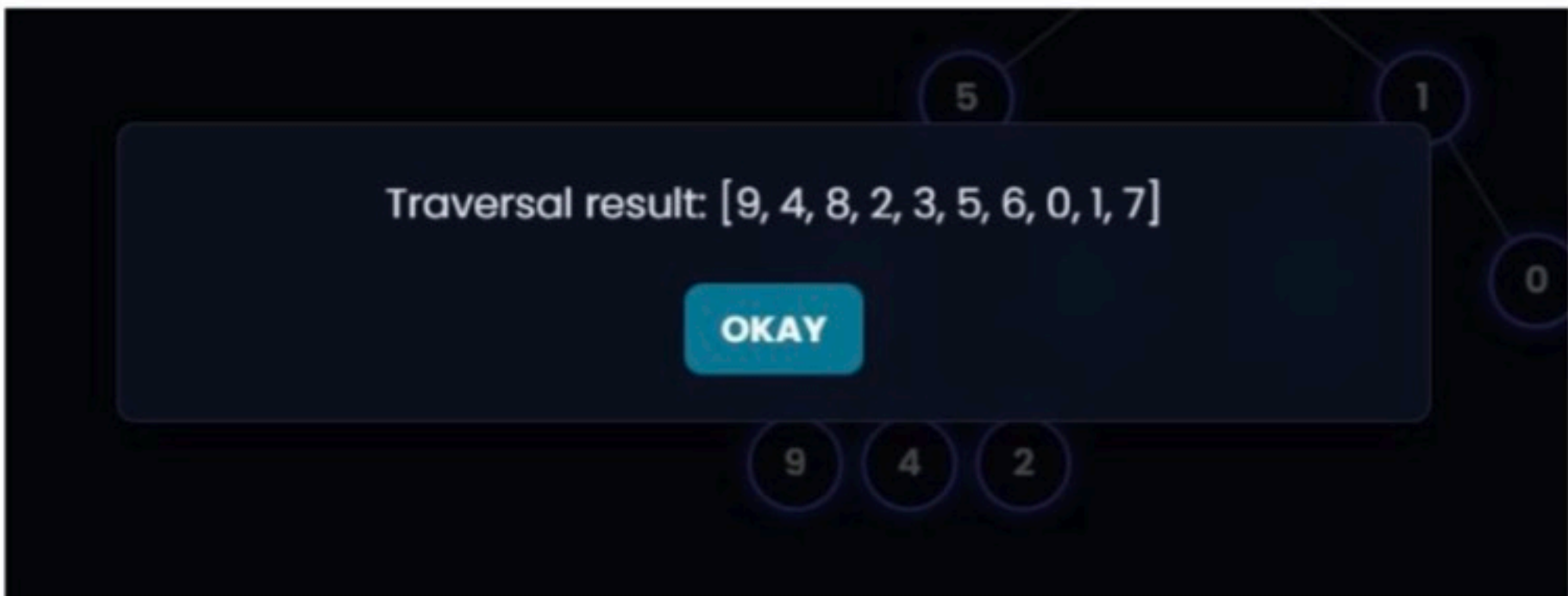
PREORDER



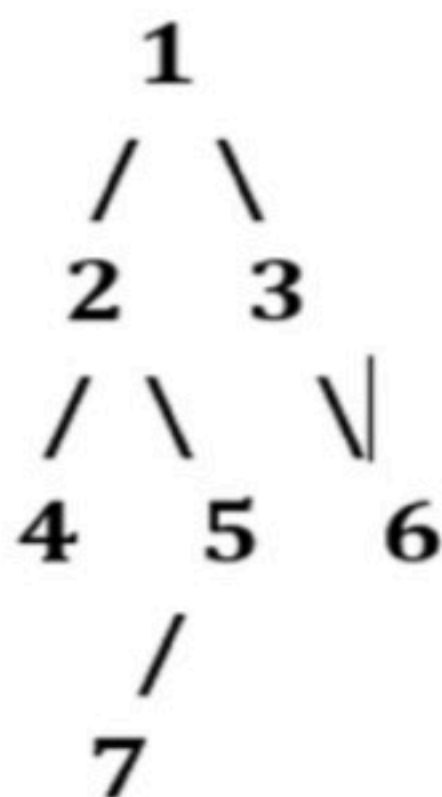
INORDER



POSTORDER



3. Write the Inorder, Preorder & Postorder traversal sequences for following trees



PREORDER

Traversal result: [4, 5, 2, 6, 7, 3, 1]

OKAY

INORDER

Traversal result: [4, 2, 5, 1, 6, 3, 7]

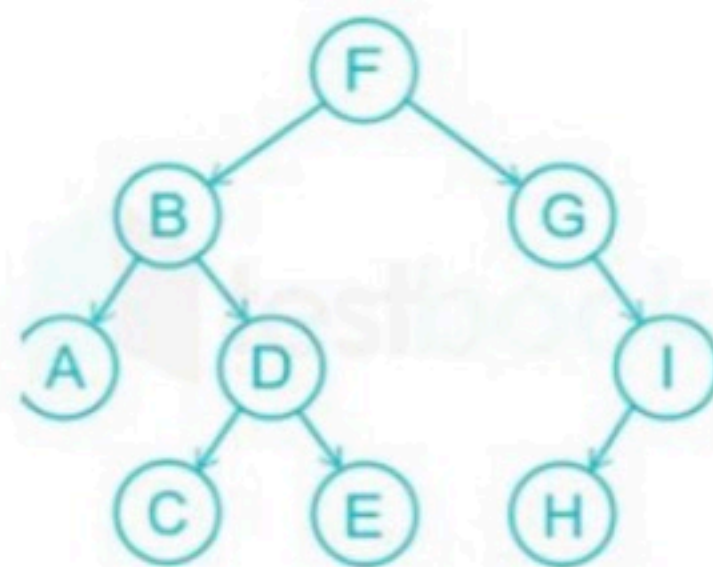
OKAY

POSTORDER

Traversal result: [4, 5, 2, 6, 7, 3, 1]

OKAY

4. What is the sequence of nodes when applying in-order traversal on the binary tree given below?



1. A, B, C, D, E, F, I, H, G

2. A, C, D, E, B, F, G, H, I

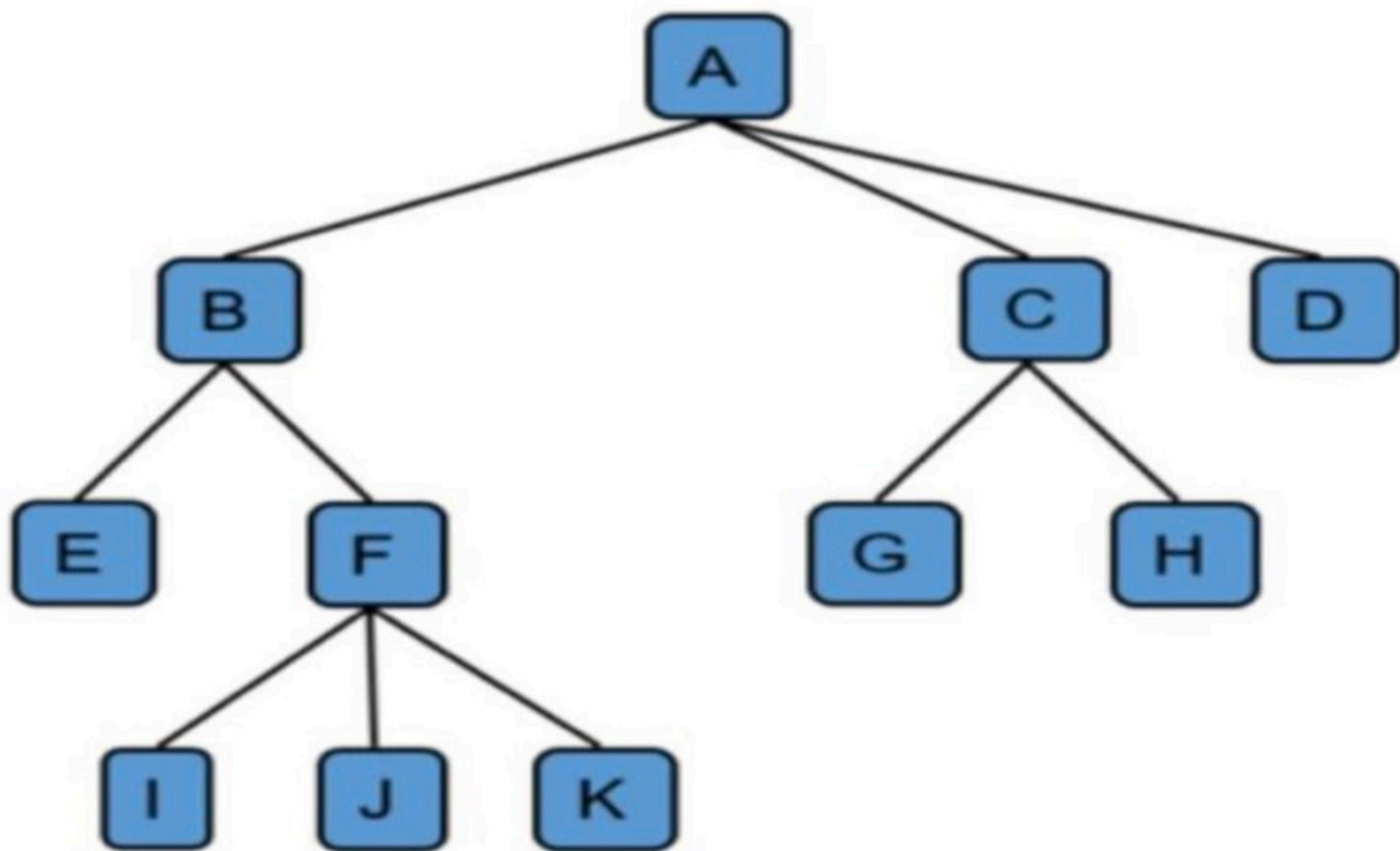
3. A, B, C, D, E, F, G, H, I

4. I, H, G, F, E, D, C, B, A

✓ Inorder Result:

A → B → C → D → E → F → G → H → I

5.



PREORDER

A → B → E → F → I → J → K → C → G → H → D

POSTORDER

$E \rightarrow I \rightarrow J \rightarrow K \rightarrow F \rightarrow B \rightarrow G \rightarrow H \rightarrow C \rightarrow D \rightarrow A$

INORDER

The tree is **not a binary tree** because:

- A has **3 children**: B, C, D
- F has **3 children**: I, J, K

So, **standard inorder traversal is NOT defined for this tree**. Inorder is normally used for a **binary tree**, where every node has at most 2 children.